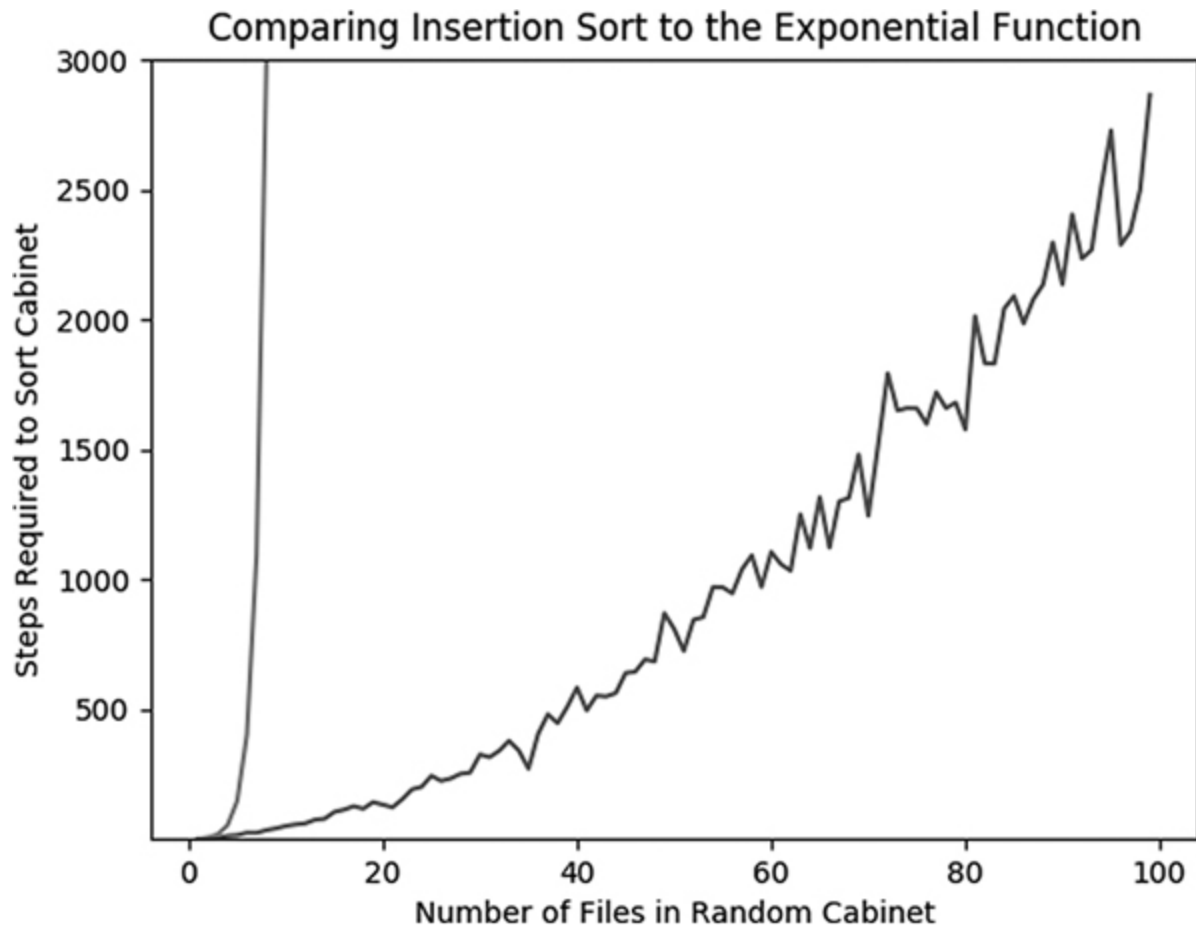




[Figure 4-2:](#) Insertion sort steps compared to the exponential function

We can see the true exponential growth curve shooting up toward infinity on the left side of the plot. Though the insertion sort step curve grows at an accelerating rate, its acceleration does not seem to get close to matching true exponential growth. If you plot other curves whose growth rate could also be called exponential, 2^x or 10^x , you'll see that all of these types of curves also grow much faster than our insertion sort step counter curve does. So if the insertion sort step curve doesn't match exponential growth, what kind of growth might it match? Let's try to plot a few more functions on the same plot. Here, we'll plot $y = x$, $y = x^{1.5}$, $y = x^2$, and $y = x^3$ along with the insertion sort step curve.

```
random.seed(5040)
xs = list(range(1,100))
ys = [check_steps(x) for x in xs]
xs_exp = [math.exp(x) for x in xs]
```